CO3-AT3

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1. What is the primary goal of Natural Language Processing (NLP)?
A) To increase computer hardware speed
B) To enable computers to process and understand human language
C) To design operating systems
D) To manage computer networks

Answer: B) To enable computers to process and understand human language

2. Which model is commonly used to describe whether a string belongs to a particular language?
A) Database model
B) Mathematical model
C) Financial model
D) Hardware model

Answer: B) Mathematical model

3. Which symbol is commonly used in regular expressions to represent zero or more occurrences of the preceding element?
A) +
B) ?
C) *
D) ^

Answer: C) *

4. Which regular expression represents a sequence containing one or more occurrences of a?
A) a*
B) a+
C) a?
D) a|

Answer: B) a+

5. What is the main function of a Finite State Automaton (FSA)?
A) To store large databases
B) To recognize patterns in strings
C) To translate programming languages
D) To generate images

Answer: B) To recognize patterns in strings

6. Which type of morphology deals with grammatical variations such as tense and number?

A) Derivational morphology
B) Inflectional morphology

C) Computational morphology
D) Structural morphology

Answer: B) Inflectional morphology

7. Which example represents derivational morphology?
A) cat → cats
B) walk → walked
C) happy → happiness
D) book → books

Answer: C) happy → happiness

8. What is the purpose of finite-state morphological parsing?
A) To identify the internal morphological structure of words
B) To identify network protocols
C) To calculate sentence length
D) To generate numerical data

Answer: A) To identify the internal morphological structure of words

9. What does the Porter Stemmer primarily perform?
A) Sentence parsing
B) Word stemming
C) POS tagging
D) Speech recognition

Answer: B) Word stemming

10. The word "unhappiness" can be analyzed as un + happy + ness. Which process is illustrated by this analysis?
A) Inflectional morphology only
B) Derivational morphology
C) Syntax parsing
D) Regular expression matching

Answer: B) Derivational morphology

11. What does an N-gram model estimate?
A) Probability of a word based on previous words
B) Number of letters in a word
C) Meaning of a paragraph
D) Number of sentences in a document

Answer: A) Probability of a word based on previous words

12. A bigram model considers how many words at a time?
A) One
B) Two
C) Three
D) Four

Answer: B) Two

13. What is the main problem with an unsmoothed N-gram model?
A) It cannot count words
B) It assigns zero probability to unseen N-grams
C) It cannot recognize nouns
D) It cannot process sentences

Answer: B) It assigns zero probability to unseen N-grams

14. What is the primary purpose of smoothing in language models?
A) To remove punctuation
B) To assign probabilities to unseen events
C) To identify verbs
D) To shorten sentences

Answer: B) To assign probabilities to unseen events

15. What does entropy measure in an N-gram language model?
A) Linguistic uncertainty
B) Number of nouns
C) Sentence length
D) Number of characters

Answer: A) Linguistic uncertainty

16. Which word class does "quickly" belong to in the sentence "She runs quickly"?
A) Noun
B) Verb
C) Adjective
D) Adverb

Answer: D) Adverb

17. What is the purpose of a tagset in POS tagging?
A) To define the set of grammatical labels used for words
B) To store sentences
C) To generate regular expressions
D) To calculate entropy

Answer: A) To define the set of grammatical labels used for words

18. In the sentence “Book a ticket,” what is the POS of “Book”?
A) Noun
B) Verb
C) Adjective
D) Adverb

Answer: B) Verb

19. Which POS tagging approach uses manually written linguistic rules?
A) Stochastic tagging
B) Rule-based tagging
C) Transformation-based tagging
D) Random tagging

Answer: B) Rule-based tagging

20. Which POS tagging method combines an initial tagging with a sequence of corrective transformations?
A) Rule-based tagging
B) Stochastic tagging
C) Transformation-based tagging
D) Dictionary-based tagging

Answer: C) Transformation-based tagging

21. What is the primary purpose of a Context-Free Grammar (CFG) in NLP?
A) To represent syntactic structure of sentences
B) To calculate word frequency
C) To identify word meanings
D) To translate documents

Answer: A) To represent syntactic structure of sentences

22. Which component of a CFG specifies how a non-terminal can be expanded?
A) Terminal symbol
B) Production rule
C) Probability table
D) Lexicon only

Answer: B) Production rule

23. In the rule S → NP VP, what does S usually represent?
A) Sentence
B) Subject

C) Semantics
D) String

Answer: A) Sentence

24. Which structure visually represents the syntactic organization of a sentence?
A) Bar chart
B) Parse tree
C) Flowchart
D) Frequency table

Answer: B) Parse tree

25. What does grammatical agreement ensure?
A) Words have the same number of letters
B) Related words have compatible grammatical features
C) All words have the same meaning
D) Sentences contain equal numbers of words

Answer: B) Related words have compatible grammatical features

26. What does subcategorization describe?
A) The arguments or complements that a word, especially a verb, requires
B) The number of letters in a word
C) The probability of a sentence
D) The meaning of punctuation

Answer: A) The arguments or complements that a word, especially a verb, requires

27. What is parsing in NLP?
A) Converting text into numbers only
B) Determining the syntactic structure of a sentence
C) Removing stop words
D) Translating words between languages

Answer: B) Determining the syntactic structure of a sentence

28. Which parsing strategy begins with the start symbol and attempts to generate the input sentence?
A) Bottom-up parsing
B) Top-down parsing
C) Stochastic tagging
D) Semantic parsing

Answer: B) Top-down parsing

29. Which parsing algorithm can parse sentences using dynamic programming and handle left-recursive grammars?
A) Porter algorithm
B) Earley parser
C) Viterbi tagger
D) PageRank algorithm

Answer: B) Earley parser

30. What is the purpose of a Probabilistic Context-Free Grammar (PCFG)?
A) To associate probabilities with grammar productions
B) To remove grammar rules
C) To identify word stems
D) To classify documents

Answer: A) To associate probabilities with grammar productions

31. What is the main objective of semantic analysis?
A) To determine the meaning of linguistic expressions
B) To count characters
C) To identify punctuation
D) To format documents

Answer: A) To determine the meaning of linguistic expressions

32. Which formal system is commonly used to represent relationships between entities and properties?
A) First-Order Predicate Calculus
B) Regular Expression
C) Finite Automaton
D) N-gram model

Answer: A) First-Order Predicate Calculus

33. In first-order predicate logic, what does P(x) typically represent?
A) A probability
B) A predicate applied to an argument
C) A parsing rule
D) A POS tag

Answer: B) A predicate applied to an argument

34. Which representation can express the statement “John loves Mary” as a relationship between entities?
A) loves(John, Mary)

B) John + Mary
C) John/Mary
D) John = Mary

Answer: A) loves(John, Mary)

35. What is syntax-driven semantic analysis?
A) Deriving meaning representations based on syntactic structure
B) Counting syntactic words
C) Removing grammatical rules
D) Generating random meanings

Answer: A) Deriving meaning representations based on syntactic structure

36. What is a lexeme?
A) An abstract vocabulary unit representing related word forms
B) A punctuation mark
C) A sentence structure
D) A grammar rule

Answer: A) An abstract vocabulary unit representing related word forms

37. What does word sense refer to?
A) The grammatical category of a word
B) A particular meaning of a word
C) The number of letters in a word
D) The pronunciation of a word

Answer: B) A particular meaning of a word

38. What is the main goal of Word Sense Disambiguation (WSD)?
A) To determine the intended meaning of an ambiguous word
B) To identify sentence boundaries
C) To count words
D) To remove prefixes

Answer: A) To determine the intended meaning of an ambiguous word

39. In "The fisherman sat on the bank," which meaning of "bank" is most appropriate?
A) Financial institution
B) River side
C) Storage facility
D) Computer memory

Answer: B) River side

40. What is the primary goal of Information Retrieval (IR)?
A) To retrieve relevant information from a collection of documents
B) To generate grammar rules
C) To identify speech sounds
D) To stem every word

Answer: A) To retrieve relevant information from a collection of documents

41. What does discourse analysis primarily study?
A) Language beyond individual sentences and its context
B) Individual characters only
C) Computer hardware
D) Word spelling only

Answer: A) Language beyond individual sentences and its context

42. What is reference resolution?
A) Determining what an expression such as "he" or "it" refers to
B) Finding the number of words in a sentence
C) Identifying punctuation marks
D) Translating a sentence

Answer: A) Determining what an expression such as "he" or "it" refers to

43. In the sentence "Ravi bought a car. He likes it," what does "He" refer to?
A) Car
B) Ravi
C) Likes
D) It

Answer: B) Ravi

44. What does text coherence refer to?
A) The meaningful and logical relationship among parts of a text
B) The number of words in a paragraph
C) The spelling of words
D) The number of punctuation marks

Answer: A) The meaningful and logical relationship among parts of a text

45. What is a discourse act or dialogue act?
A) The communicative function performed by an utterance
B) The number of words in dialogue
C) A grammatical error
D) A punctuation rule

Answer: A) The communicative function performed by an utterance

46. Which of the following is an example of a dialogue act?
A) Request
B) Noun
C) Prefix
D) Stem

Answer: A) Request

47. What is the main purpose of a conversational agent?
A) To interact with users through natural language
B) To perform only mathematical calculations
C) To store images
D) To compile source code

Answer: A) To interact with users through natural language

48. What is the purpose of surface realization in language generation?
A) To convert an abstract representation into natural-language text
B) To identify word senses
C) To parse a sentence
D) To calculate N-gram entropy

Answer: A) To convert an abstract representation into natural-language text

49. Which machine translation approach uses an intermediate language-independent representation?
A) Direct translation
B) Interlingua
C) Word-for-word translation
D) Rule deletion

Answer: B) Interlingua

50. What do statistical approaches to machine translation primarily learn from?
A) Parallel and bilingual language data
B) Only punctuation marks
C) Computer hardware specifications
D) Regular expressions alone

Answer: A) Parallel and bilingual language data